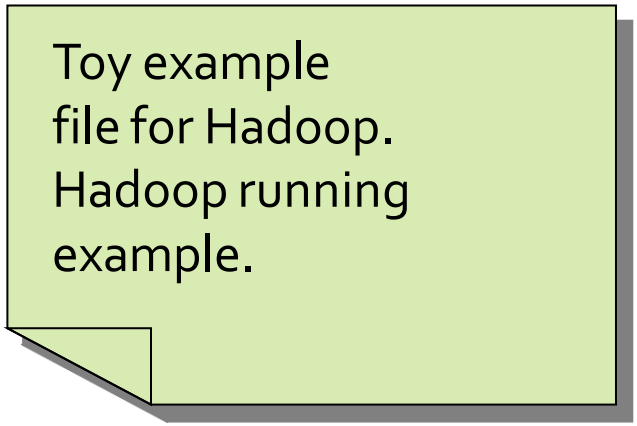
Exercise #9

- Word count problem
  - Input: (unstructured) textual file
  - Output: number of occurrences of each word appearing in the input file
- Solve the problem by using in-mapper combiners

Exercise #9 - Example

- Input file



- Output pairs

(toy, 1)
(example, 2)
(file, 1)
(for, 1)
(hadoop, 2)
(running, 1)

Exercise #12

- Select outliers
  - Input: a collection of (structured) textual files containing the daily value of PM10 for a set of sensors
    - Each line of the files has the following format

      sensorId**,**date**\t**PM10 value (µg/m^3)**\n**
  - Output: the records with a PM10 value below a user provided threshold (the threshold is an argument of the program)

Exercise #12 - Example

- Input file

```
s1,2016-01-01    20.5
s2,2016-01-01    60.2
s1,2016-01-02    30.1
s2,2016-01-02    20.4
s1,2016-01-03    55.5
s2,2016-01-03    52.5
```

  - Threshold: 21
- Output

```
s1,2016-01-01    20.5
s2,2016-01-02    20.4
```